Gauss Elimination with pivoting Techniques

Disadvantage :-

1) Division by zero :-

$$\left[\begin{array}{ccc|c} 0 & 3 & 4 & 2 \\ 2 & -1 & 2 & 3 \\ \cdot & & & \end{array}\right] \qquad a_{11} = 0$$

we can not do the row operation.

2) Round OFF errors :-

suppose our equation is

$$10^{-5} x_1 + 3x_2 + 4x_3 = 2$$
$$x_1 - x_2 + x_3 = 3$$

Number is very small.

$$\begin{bmatrix} 0 & 2 & 1 \\ 1 & 4 & 3 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

In here pivot Number is zero so.

we can not do the row operation

then.

we can do Interchanging row

$$\begin{bmatrix} 1 & 4 & 3 \\ 0 & 2 & 1 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} y \\ x \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

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then we do the row operation

$$\left(\begin{array}{ccc|c} 1 & 4 & 3 & 1 \\ 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 2 \end{array}\right)$$

$R_3 - 2R_1 \rightarrow R_3$

$$\begin{array}{cccc} 2 & 2 & 1 & 2 \\ 2 & 8 & 6 & 2 \\ \hline 0 & -6 & -5 & 0 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 4 & 3 & 1 \\ 0 & 2 & 1 & 2 \\ 0 & -6 & -5 & 0 \end{array}\right)$$

$R_3 - \left(-\frac{6}{2}\right)R_2 \rightarrow R_3$

$$\begin{array}{cccc} 0 & -6 & -5 & 0 \\ 0 & 6 & 6/2 & 6 \\ \hline 0 & 0 & -2 & 6 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 4 & 3 & 1 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & -2 & 6 \end{array}\right)$$

the we do the Back substitution.